DIONIS LUDOVICK NASUWA

Data Analyst & Software Developer | Relational Databases, Fraud Analytics & Systems Infrastructure

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PROFESSIONAL SUMMARY

Computer Science graduate from the University of Dar es Salaam (CoICT) and Data Analyst at Halotel headquarters. Focuses on querying large relational databases, cleaning operational datasets, detecting transaction anomalies, and delivering clear Tableau reports for management. Uses full-stack software development as a practical foundation to design resilient backend services, automate data pipelines, and maintain strict data integrity standards.

EDUCATION & CERTIFICATIONS

BSc in Computer Science: University of Dar es Salaam (CoICT) — 2023 - 2026

Data Analytics Essentials: Cisco Networking Academy — Sept 2026

Full Stack Open Certification (Grade 5/5 Maximum Score): University of Helsinki — July 2026

GDPR Foundations Certification: Measured Collective — Sept 2026

Key Coursework: Database Management Systems (SQL), Data Structures & Algorithms, Distributed Systems, Operating Systems & Linux Administration, Computer Networks, Systems Programming.

TECHNICAL EXPERIENCE

Data Analyst (HQ) • **Halotel (Viettel Tanzania Plc)** — Sept 2026 - Present | Dar es Salaam

- Configure and declare Value Added Services (VAS) for SMS shortcodes, querying PostgreSQL and Oracle databases with SQL to audit service delivery logs and resolve routing discrepancies.
- Provide technical support and written guidance to external corporate partners, monitoring message traffic trends and building Tableau and Excel dashboards for operational reporting.

Co-Founder & Technical Consultant • **DigitalOps (Independent Consultancy)** — Jan 2026 - Present | Dar es Salaam

- Architect backend systems and payment integrations for East African commercial clients, using PostgreSQL advisory locks and retry queues to prevent duplicate transaction crediting.
- Author technical proposals, system specifications, and client reports to guide deployment timelines and operational risk mitigation.

Software Engineering Intern • **Mainstream Group (Technology Division)** — March 2026 - May 2026 | Dar es Salaam

- Engineered modular backend services in TypeScript and Node.js for multi-module applications, optimizing SQL queries to improve data retrieval speeds.
- Participated in bi-weekly agile sprint reviews, documented API integration specifications, and conducted peer code reviews across release pipelines.

Software Development Trainee • **UDICTI / StartHub Africa Partnership** — July 2025 - Sept 2025 | Dar es Salaam

- Built cross-platform components for the Smart Cargo delivery application using Flutter, connected to a Django and PostgreSQL backend with bilingual Swahili and English support.
- Modeled multi-factor pricing calculations in Excel and Python based on distance matrices and weight brackets, presenting milestone reports to project supervisors.

IT Infrastructure Trainee • **TPC Limited** — July 2024 - Sept 2024 | Kilimanjaro

- Configured managed network switches, routers, and static IP bindings to maintain local area network continuity, diagnosing hardware and operating system issues across departmental workstations.

SELECTED TECHNICAL PROJECTS

PaySim Mobile Money Analytics: Financial Fraud Forensics & Dashboarding — 2026

PostgreSQL · SQL · Tableau Public

- Analyzed 6.36 million transaction records in PostgreSQL using CTEs and window functions, discovering that static threshold rules missed 99.8 percent of fraudulent transfers.
- Published an interactive three-tab dashboard on Tableau Public displaying system volume trends, hourly fraud occurrences, and balance discrepancy heatmaps.

Paydable Engine: Payment Idempotency & Webhook Infrastructure — 2026

Node.js · PostgreSQL · Redis · Distributed Locks

- Engineered a payment integration service using PostgreSQL advisory locks to guarantee transaction idempotency against duplicate mobile money webhooks.
- Implemented background retry queues with exponential backoff intervals across Selcom, AzamPay, and Stripe callbacks to ensure reliable state reconciliation.

Plaqode Platform: Distributed Monorepo & Identity Architecture — 2026

TypeScript · Next.js · Fastify · Turborepo · PostgreSQL · Docker

- Architected an eight-application Turborepo monorepo with isolated PostgreSQL schemas per service domain, preventing schema coupling across ticketing and identity modules.
- Designed a stateless RS256 JWT authentication service with Fastify, enabling independent public-key signature verification across client applications.

TECHNICAL SKILLS & COMPETENCIES

Data & Analytics: SQL (PostgreSQL, OracleDB), Tableau, Microsoft Excel (pivot tables, modeling), Python, Data Cleaning, Transaction Fraud Forensics, Database Normalization.

Software Engineering: TypeScript, JavaScript, Node.js, Fastify, React, Next.js, Django, Flutter, RESTful APIs, RS256 Authentication.

Infrastructure & Tools: Docker, Git, Linux (Ubuntu/Debian), PostgreSQL, Redis, Turborepo, Microsoft Office Suite (Word, PowerPoint, Excel).

Methodologies & Standards: Agile/Scrum, Data Governance & Privacy (GDPR), Database Modeling, Code Review, Technical Documentation.